Resume

Pengbo Sun

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Pengbo Sun
Computer Vision and Robotic Engineer

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Resume

Date of birth: 13.10.1995 Place of birth: Beijing

Educational Experience

10.2019–02.2022
Master Electrical Engineering: Automation | Technical University Berlin

GPA: 1.7
Study content: Machine Learning, Robotics, Computer Vision
Project: ORB-SLAM, Gaze Tracking
Masterthesis: Deep learning based 6D pose estimation of general objects

10.2016–10.2019
Bachelor Electrical Engineering: Automation | Technical University Chemnitz

GPA: 1.7
Study content: Control Theory, Robotics, Sensor Signal Processing
Project: Mobile Robotics, Image Segmentation
Bachelorthesis: Next best view plannig for industrial robots

Work Experience

Since 04.2022
Computer Vision, Robotic Engineer | Agile Robots AG

Project: IPA Project

Technical Manager
Motion Planning and Simulation Implementation and Development.
PointCloud based AMR localization algorithm Developement.
Development of a robust pose estimator with multimodalities.
Integration and Deployment in China
C++ 17, OpenCV, Point Cloud Library (PCL), OpenMP

Project: Humanoid Robot

Leading Developer of Robot Vision Module:
Robot Self-Calibration: Developed a vision-based approach enabling the robot to self-calibrate its arms and hands without the need for external patterns.
Object Pose Estimation: Created a point cloud-based pose estimator that is both fast and robust against outliers.
Object Tracking: Developed an optimization-based object tracker utilizing RGB-D images.
C++ 17, Python, OpenCV, Point Cloud Library (PCL) , Open3d

Project: Rcam Cowling - Iphone camera assembly task

Leading Developer of Computer Vision and Integration Module :
Object Pose Estimation: Developed keypoint and shape based template matching algorithm , which is used to detect the pose of camera cowling and housing.
Integration Task: Integrated the assenbly work flow with TCP/IP communication.
C++ 17, OpenCV.

Project: Agile Core - An intelligent robot programming platform

Main Developer of Computer Vision Module:

2D Segmentation: developed deep learning based mask segmentation approach
Grap pose Estimation: developed point cloud based grasp pose estimation approach

Python, Pytorch, Open3d

Intership Experience

10.2021–04.2022

Masterthesis: Deep learning based 6D pose estimation of general objects | Agile Robots AG

Object pose can be described with a 6D vector. In my Masterthesis, i developed an augumented autoencoder, which is able to extract geometric feature from RGB images. The autoencoder is trained with objects images rendered in different background. In order to generalize the deep model, multipath learning strategy is applied to the approach. The proposed approach is trained and tested with T-Less Dataset.

11.2019–03.2020

Internship: Deep Learning Engineer | Audi China

During my internship in Audi China, i worked on Driveablility Projects, which aims to predict the driveability of vehicles with gas pedal and accleration signals. In order to filter noise from the given dataset, i applied a CNN sturcture to this task.

Programming Languages:

C/C++

Python

Matlab/Simulink

Framework and Tools:

Conan

JIRA

Confluence

Gitlab

Makefile

OpenCV

Open3D

PCL

PyTorch

Tensorflow

Docker

Conda

Languages

Chinese: native speaker

English: C1

German: C1

Soft Skills

Creativity

Target focus

Analylitical thinking

Teamwork

Communication

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简历

生日：1995.10.13　出生地：北京

教育背景

2019.10 – 2022.02
硕士专业：电气工程 – 自动化方向 | 柏林工业大学
绩点：1.7
学习内容：机器学习、机器人技术、计算机视觉
项目经历：ORB-SLAM、视线追踪
硕士论文：基于深度学习的一般物体的6D位姿估计

2016.10 – 2019.10
学士专业：电气工程 – 自动化方向 | 开姆尼茨工业大学
绩点：1.7
学习内容：控制理论、机器人技术、传感器信号处理
项目经历：移动机器人、图像分割
学士论文：工业机器人的下一最佳视角规划

工作经历

2022.04 至今
计算机视觉与机器人工程师 | Agile Robots AG

项目：IPA 项目

技术负责人
运动规划与仿真模块的实现与开发
基于点云的自主移动机器人（AMR）定位算法开发
多模态鲁棒位姿估计算法开发
中国本地化集成与部署
技术栈：C++17，OpenCV，点云库（PCL），OpenMP

项目：人型机器人

机器人视觉模块主力开发者
机器人自标定：开发了一种基于视觉的方法，使机器人无需外部标定板即可自主标定其手臂与手部位置
物体位姿估计：实现了一种基于点云的位姿估计器，具备高速和强抗离群点能力
物体追踪：开发了基于RGB-D图像的优化型目标追踪器
技术栈：C++17，Python，OpenCV，点云库（PCL），Open3D

项目：Rcam Cowling – iPhone 摄像头组件装配任务

计算机视觉与集成模块负责人
物体位姿估计：开发了结合关键点和形状模板匹配的算法，用于检测摄像头护罩与外壳的空间位姿
集成任务：将装配流程与 TCP/IP 通信接口进行集成
技术栈：C++17，OpenCV

项目：Agile Core - 智能机器人编程平台

2D 分割：开发了基于深度学习的掩码分割方法
抓取位姿估计：实现了基于点云的抓取位姿估计算法
技术栈：Python，PyTorch，Open3D

实习经历

2021.10 - 2022.04

硕士论文：基于深度学习的一般物体的6D位姿估计 | Agile Robots AG

物体的姿态可以通过一个六维向量进行描述。在我的硕士论文中，我开发了一种增强型自动编码器（Augmented Autoencoder），能够从RGB图像中提取几何特征。该自动编码器在不同背景下渲染的物体图像上进行训练。为了提升模型的泛化能力，我引入了多路径学习策略（Multipath Learning Strategy）所提出的方法在 T-Less 数据集上进行了训练与测试。

2019.11 - 2020.03

实习经历：深度学习工程师 | 奥迪中国（Audi China）

在奥迪中国的实习期间，我参与了可驾驶性预测项目（Driveability Projects），该项目旨在通过油门踏板信号与加速度信号预测车辆的可驾驶性。为了从原始数据中去除噪声，我将卷积神经网络（CNN）结构应用于该任务中，提升了数据处理与预测的鲁棒性。

编程语言:

C/C++

Python

Matlab/Simulink

专业技能:

Conan

JIRA

Confluence

Gitlab

Makefile

OpenCV

Open3D

PCL

PyTorch

Tensorflow

Docker

Conda

语言技能

中文：母语
德语：C1（高级流利程度）
英语：C1（高级流利程度）

综合能力

创造力

专注力

分析能力

团队合作力

沟通能力